YData synthetic data generator

Synthetic Data Quality Report

YData

PRIVACY LEVEL

HIGH FIDELITY



Fidelity Score

890

REAL DATA RECORDS



Utility Score

22

SYNTHETIC DATA RECORDS GENERATED

7

COLUMNS



Privacy Score

0

ANONYMIZED COLUMNS

FIDELITY SCORES



Fidelity Score

Fidelity measures how well the synthetic data statistically matches the original records. It is provided through univariate and multivariate metrics, model and assumption-free. The anonymized columns are not considered for the fidelity scores since they are necessarily different between the real and synthetic records in order to achieve anonymization.

STATISTICS DISTANCE

The metrics below provide summarized information of the detailed statistical metrics calculated through YData's profiling.

0.96	0.88	1.00
CORRELATION SIMILARITY	DISTANCE DISTRIBUTION	SYNTHETIC CLASSIFIER

The **CORRELATION SIMILARITY** measures how close are synthetic and real correlation matrices. It is bounded between [0-1] metric, the closer to 1, the higher fidelity.

DISTANCE DISTRIBUTION measures the features' distribution similarity between original and generated data. The Chi-squared test evaluates features with discrete distributions, and the Kolmogorov-Smirnov test evaluates features with continuous distributions. Returns values between [0, 1], closer to 1 is desirable.

The **SYNTHETIC CLASSIFIER** score is the ROC-AUC of a model trained to distinguish between real and synthetic data. A score close to 1 indicates that the estimator is not able to discriminate between the original and synthetic data records, making the data safer to be shared. This metric is bounded between [0-1].

MISSING VALUES SIMILARITY

The **MISSING VALUES SIMILARITY (MVS)** measures how close are the percentages of missing values in the synthetic and real data. This metric is bounded between [0-1], where 1 represents the same percentage of missing data. The two tables below present the five features with the highest and lowest similarity.

1.00

MISSING VALUES SIMILARITY

Feature	MVS (Highest)
age	1.0
income	1.0
credit_score	1.0
defaulted	1.0
occupation	1.0

Feature	MVS (Lowest)
credit_score	1.0
defaulted	1.0
occupation	1.0
zip_prefix	1.0
age_group	1.0

STATISTICAL SIMILARITY

The **STATISTICAL SIMILARITY** measures how similar are the synthetic and real data considering five metrics: mean, standard deviation, median, 25% quantile, and 75% quantile. Each similarity is bounded between [0-1], where 1 represents equal values. Only numerical features are considered in this analysis.

0.98
MEAN

0.98
STD. DEV.

0.98
MEDIAN

0.97
Q25%

0.97
Q75%

Feature	Mean	Std. Dev.	Median	Q25%	Q75%
age	0.99	0.98	0.98	0.96	0.95
income	0.99	0.97	0.99	0.99	1.00
credit_score	0.95	0.98	0.96	0.95	0.95

DISTRIBUTION METRICS

The distribution metrics compare the probability distributions of the real and synthetic data variables. These metrics return values between [0, 1], where closer to 1 is desirable (i.e., the distributions are likely the same).

KOLMOGOROV-SMIRNOV TEST

The **KOLMOGOROV-SMIRNOV (KS) TEST** compares the distribution between two continuous variables (real and synthetic data) using the empirical CDF. The two tables below present the five features with the highest and lowest values for this test.

0.85

KOLMOGOROV-SMIRNOV TEST

Feature	KS Test (Highest)
age	0.89
income	0.84
credit_score	0.83

Feature	KS Test (Lowest)
age	0.89
income	0.84
credit_score	0.83

TOTAL VARIATION DISTANCE

The **Total Variation Distance (TVD)** compares the marginal distribution between two categorical variables (real and synthetic data). The two tables below present the five features with the highest and lowest values for this metric.

0.81

TOTAL VARIATION DISTANCE

Feature	TVD (Highest)
age_group	0.86
defaulted	0.84
zip_prefix	0.78
occupation	0.76

Feature	TVD (Lowest)
age_group	0.86
defaulted	0.84
zip_prefix	0.78
occupation	0.76

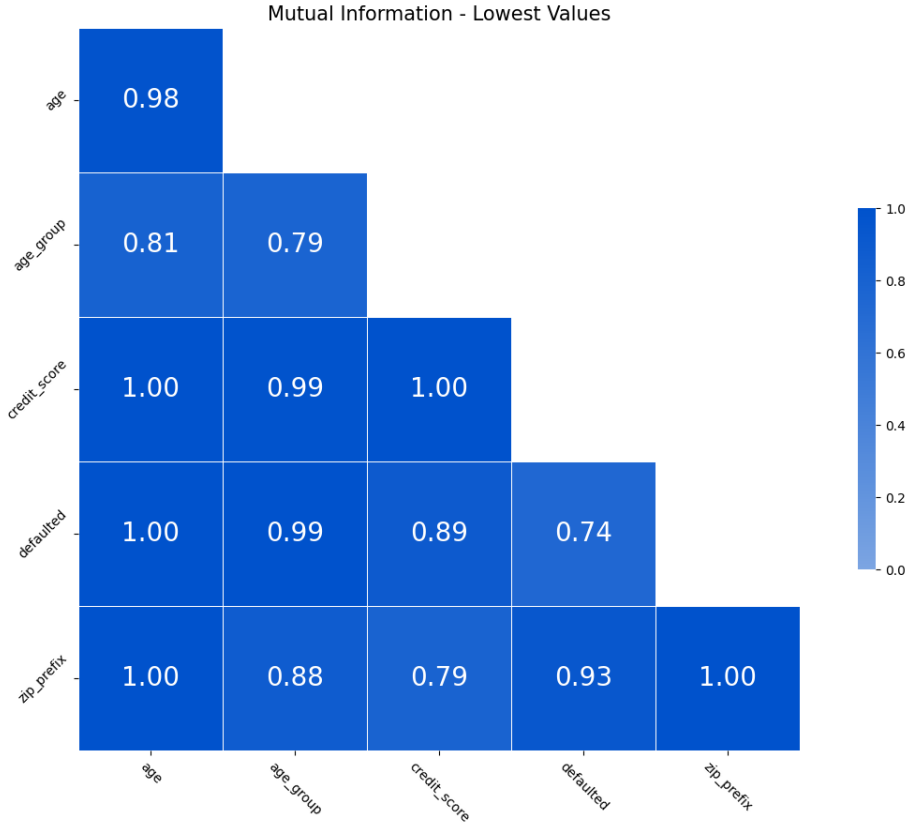
MUTUAL INFORMATION

MUTUAL INFORMATION (MI) measures how much information can be obtained about one feature by observing another. This metric calculates the similarity between real and synthetic MI values for each pair of features. It returns values between [0, 1], where closer to 1 is desirable (i.e., equal MI).

0.94

MUTUAL INFORMATION





COVERAGE METRICS

The coverage metrics describe how well the real data variables are represented in the synthetic data and they return values between [0, 1], where 1 represents complete coverage. These metrics are divided into two groups:
• Metrics for categorical data, which include **Category Coverage** and **Missing Category Coverage**.
• Metrics for numerical data, namely the **Range Coverage**.

CATEGORICAL DATA

The following metrics are specific to categorical data.

CATEGORY COVERAGE

The **CATEGORY COVERAGE (CC)** computes the ratio of real data categories that are represented in the synthetic data. The two tables below present the five features with the highest and lowest coverage.

0.84

CATEGORY COVERAGE

Feature	CC (Highest)
zip_prefix	1.00
age_group	1.00
occupation	0.88
defaulted	0.50

Feature	CC (Lowest)
zip_prefix	1.00
age_group	1.00
occupation	0.88
defaulted	0.50

MISSING CATEGORY COVERAGE

The **MISSING CATEGORY COVERAGE (MCC)** computes the similarity ratio of the cardinality between the real and synthetic categorical values. The two tables below present the five features with the highest and lowest coverage.

0.02

MISSING CATEGORY COVERAGE

Feature	MCC (Highest)
age_group	0.03
zip_prefix	0.03
occupation	0.02
defaulted	0.01

Feature	MCC (Lowest)
age_group	0.03
zip_prefix	0.03
occupation	0.02
defaulted	0.01

NUMERICAL DATA

The following metrics are specific to numerical data.

RANGE COVERAGE

The **RANGE COVERAGE (RC)** computes the similarity ratio between the numerical variables domain in the real dataset compared to the synthetic one. The two tables below present the five features with the highest and lowest coverage.

0.63

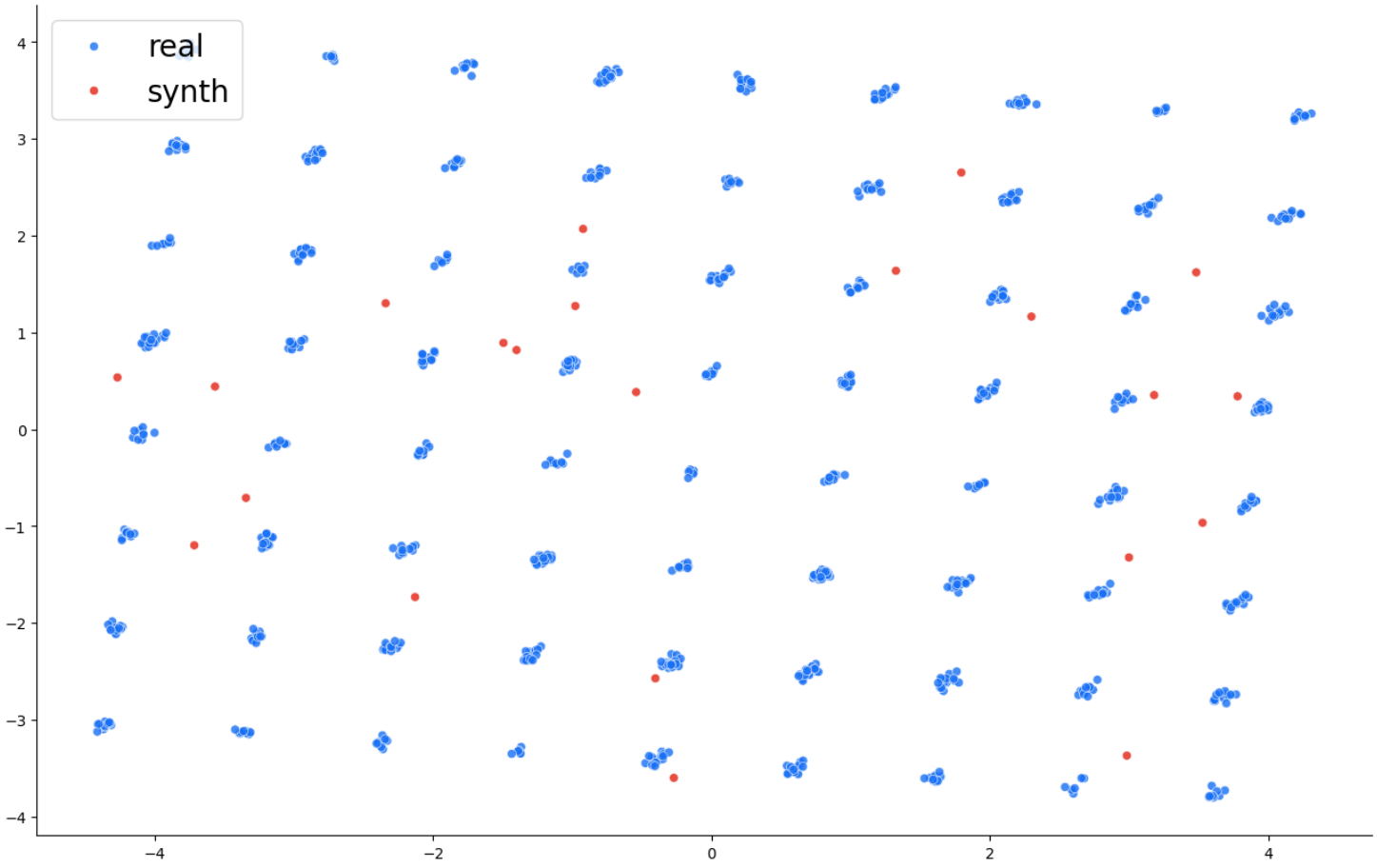
RANGE COVERAGE

Feature	RC (Highest)
age	0.94
credit_score	0.74
income	0.21

Feature	RC (Lowest)
age	0.94
credit_score	0.74
income	0.21

DIMENSIONALITY REDUCTION

The dimensionality reduction visualization plots show how closely the distribution of the synthetic data resembles that of the original data on a two-dimensional graph. Principal Component Analysis (PCA) algorithm used to reduce the datasets dimensionality.
PCA captures any fundamental difference in the distributions of the datasets. The scatterplots depict this difference visually.
The two first main Eigenvectors together explain 75.59% of the total variance of the dataset.



UTILITY SCORES



Utility Score

YData's Synthetic Data Profile utility scores are grouped into two main classes:
• Query quality enables you to understand the quality of the generated synthetic data to answer the questions with similar results as the original data.
• The Predictive performance allows you to understand the effects of an ML model trained on your synthetic data and later applied in production applications. It provides a supervised perspective of the synthetic data generated.
The anonymized columns are not considered for the utility scores since they are necessarily different between the real and synthetic records in order to achieve anonymization.

QSCORE

0.37

QSCORE

QScore measures the utility of the synthetic data by comparing the returns of random aggregation-based queries on both the synthetic and original datasets. The QScore ensures that future queries performed on the synthetic data would return the same statistical characteristics as those on the original data. Score between [0, 1]. It is recommended a QScore above 0.8 if leveraging the synthetic data for BI activities or unsupervised learning.

PREDICTION METRICS

The prediction metrics highlight the AUROC performance of several machine learning (ML) models that are trained separately on both the synthetic and original datasets. The score is calculated by testing the models on withheld data from the original dataset. The Train Synthetic Test Real (TSTR) score measures the performance of an estimator trained on synthetic data and later evaluated on real data. Train Real Test Real (TRTR) provides the estimator score expected if actual data was available. Suppose the TSTR and TRTR scores are comparable. In that case, the data generated by the synthesizer has a similar predictive performance as the original. This score is calculated using (TSTR/TRTR) and for that values between [0, +inf[are expected. Values >1 mean that the synthetic data had a better performance than the original data. For Machine Learning use, a predictive score above 0.8 is recommended.

ERROR LOGS

[Train on Synthetic Test on Real] boolean index did not match indexed array along axis 0; size of axis is 1 but size of corresponding boolean axis is 2

FEATURE IMPORTANCE

The feature importance score extends the prediction score by highlighting the differences in the importance of the most critical features for a downstream model trained on the synthetic data and the same downstream model trained on the original data. Synthetic data of excellent quality shall preserve an approximate order for variables with higher importance.

ERROR LOGS

[Feature Importance] y contains 1 class after sample_weight trimmed classes with zero weights, while a minimum of 2 classes are required.

PRIVACY SCORES



Privacy Score

Privacy indicates the level of confidentiality of the synthetic data. Larger and more complex datasets typically offer higher privacy scores and more protection against attacks.

PRIVACY METRICS

0.0	0.02
EXACT MATCHES	IDENTIFIABILITY RISK

The **EXACT MATCHES** score counts the percentage of sensitive records in the synthetic data that match the records in the original dataset. It is bounded between [0-1]. The score must be 0 for safe data-sharing.

The **IDENTIFIABILITY** score returns the re-identification risk on the real dataset from the synthetic dataset. It is bounded between [0-1]. A score of 0 represents no re-identification risk.